

轮趣科技

WHEELTEC Self-Balancing Bike Getting Started

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Version Description:

Version	date	Contents
V1.0	2022/06/23	First release

Website: www.wheeltec.net

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1. Getting Started

1.1 Use of the Car Balancing Function

After receiving the bicycle, you can start using it after confirming that the appearance is not damaged.

First, turn on the main switch of the car, and the indicator lights and display screen on the car will light up.

Two ways to start: One way is to hold the car upright on the ground with your hand and press the user button to start the car.

Another way is to put the car on the ground and push it forward to start the car.

When the tilt exceeds 40°, the rear wheel motor will automatically shut down. The main switch and user buttons of the car are shown in Figure 1-1.

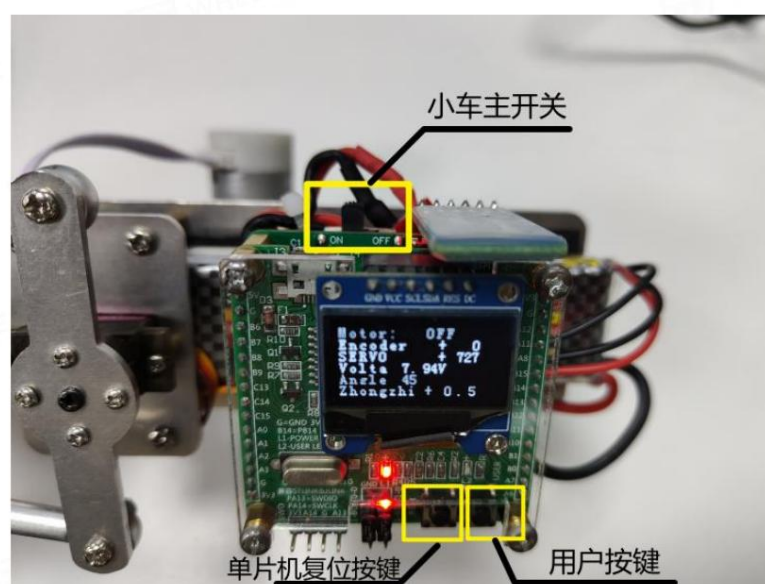


Figure 1-1 Schematic diagram of the car mainboard

1.2 Introduction to car display parameters

The car's OLED display data is as follows:

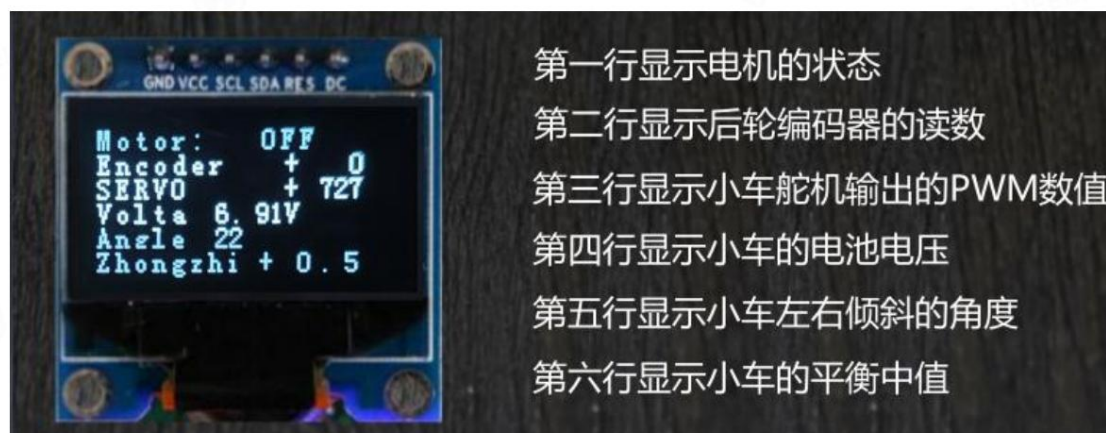


Figure 1-2 Car OLED display numerical description

1.3 Bluetooth remote control car

ÿ Control the car

Install the MiniBalance APP on your phone, search for the Bluetooth named BT04-A in the APP and connect it.

After the connection is completed, you can see the data sent back by the car in the Debug column, indicating that the connection has been successful.

At this time, put the car on the ground and start balancing, then you can control the movement of the car in the APP.

The car only supports joystick control of direction, and cannot control acceleration, deceleration, and backward movement. The control instructions are as follows:



Figure 1-3 Mobile APP interface

APP joystick ÿ	ÿ	ÿ ÿ		ÿ	ÿ	ÿ ÿ	
The state of the car remains unchanged.	Left, left, left, right, right, right						

Table 1-1 Mobile APP Operation Instructions

ÿ Use the APP to view waveforms and adjust parameters online

Click "Waveform" to enter the waveform viewing interface. The waveform interface has 3 parameters for users to view:

They are the car's angle, angular velocity and the PWM value of the input servo.

Click "Debug" to enter the adjustment interface. Before debugging, you need to click the three dots in the upper right corner of the page.

Then click Get Device Parameters to get the car's data. The parameter descriptions on the debugging interface are as follows:

The first three rows are the actual values of Kp, Ki, and Kd adjusted for LQR error; the fourth row is the actual values of Kp, Ki, and Kd adjusted for LQR error;

The mechanical median value of the body roll angle after being magnified 100 times, which can be adjusted by the potentiometer on the main board;

The 5th and 6th lines are the left and right steering ranges of the car after it is magnified 10 times. When the user adjusts the value, the input value will be

Reduce it by 10 times and then assign the value in the program.



Figure 1-4 Online parameter adjustment via mobile app

1.4 Use PC to view waveform

When the car is turned on, the APP is used by default to send data. If you need to use a PC to view the waveform,

You need to double-click the user button on the car's mainboard. Then you will see that the OLED screen stops refreshing, which means that the car

The car entered the PC host computer waveform display mode.

Before using the PC to view the waveform, you need to connect the car to the computer with a data cable. The connection instructions are as follows:

As shown in Figure 1-5. We found it in the directory "6. Software and Driver\6. MiniBalance PC Software"

MiniBalance waveform display host computer software, right click to open as administrator, select the corresponding serial port,

Select 128000 as the baud rate, then click "Open Serial Port" to see the waveform. The waveform is displayed in small

The vehicle's left and right tilt angles, the angular velocity of the left and right tilt angles, and the PWM value input to the servo are shown in Figure 1-6.



Figure 1-5 Connection prompt

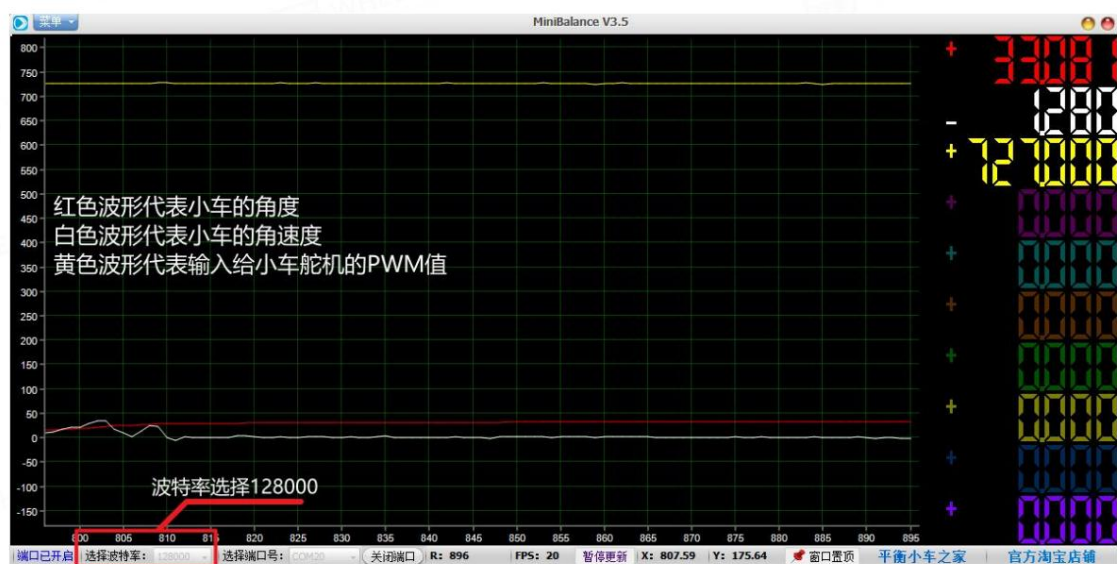


Figure 1-6 PC host computer waveform display

1.5 Car balance adjustment

If the trolley is not well balanced and falls over during operation, you can adjust the

Two potentiometers optimize the effect. Note: We will debug before delivery, and users generally do not need to adjust by themselves

Festival.

Faucet adjustment: Turn the potentiometer on the left and adjust it until the front of the car is facing straight ahead.

Balance adjustment: Place the car upright and observe the "Angle" value on the display.

When the car is placed vertically, this value is the balance center value. Turn the potentiometer on the right to adjust it to this value.

For example: If the car is placed in a vertical balance, the value displayed on the "Angle" is 1, then turn the potentiometer to

If the car is placed vertically and balanced, the value of "Angle" is 359, then turn the electric

The bit shifter adjusts the value of "Zhongzhi" to -1.

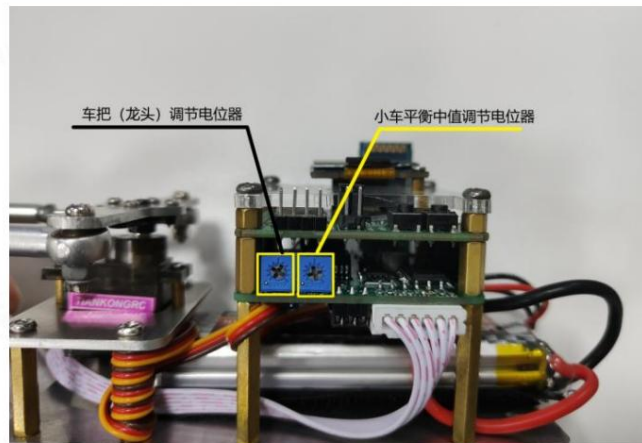


Figure 1-7 Schematic diagram of the position of the trolley potentiometer

1.6 Car Charging

The car is powered by a 2s aircraft battery. If the battery is over-discharged, it will cause permanent damage to the battery.

It is equipped with a voltage detection function. When the voltage shown on the display screen is lower than 7.1V, the motor of the car will automatically shut down.

When the voltage is lower than 7.1V, the car should be charged in time.

When charging, the two indicator lights on the charger are red and green when fully charged.

It is normal for the charger to get hot.

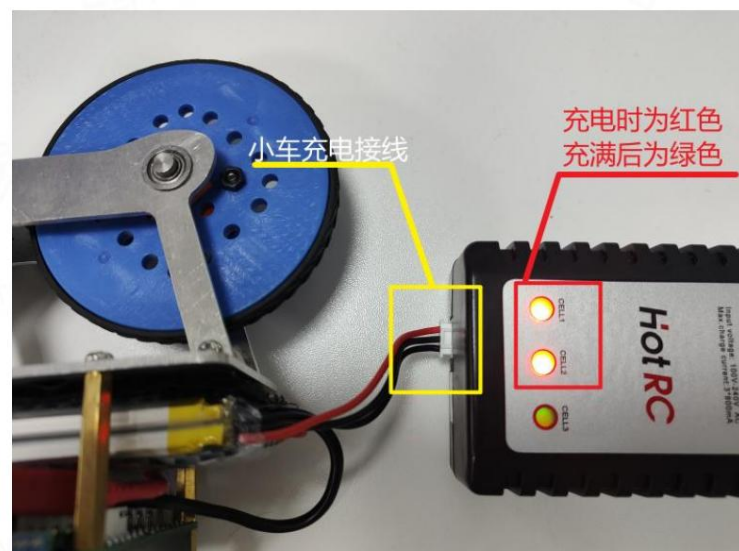


Figure 1-8 Schematic diagram of car charging

1.7 Download the program

The motherboard uses a one-key download circuit, which makes downloading programs very convenient.

Just follow the line.

ÿ Hardware preparation

1. Car mainboard

2. MicroUSB mobile phone data cable

Software preparation

1. FlyMCU burning software (included in the package)

2. The corresponding USB to TTL module CH340G driver (driver is also provided in the attached document)

After the driver is successfully installed, you can open the device manager to view



Wiring

It's very simple, just connect the computer and the board with a data cable.

FlyMcu software settings, open the FlyMcu software in the attached materials, and make the following settings

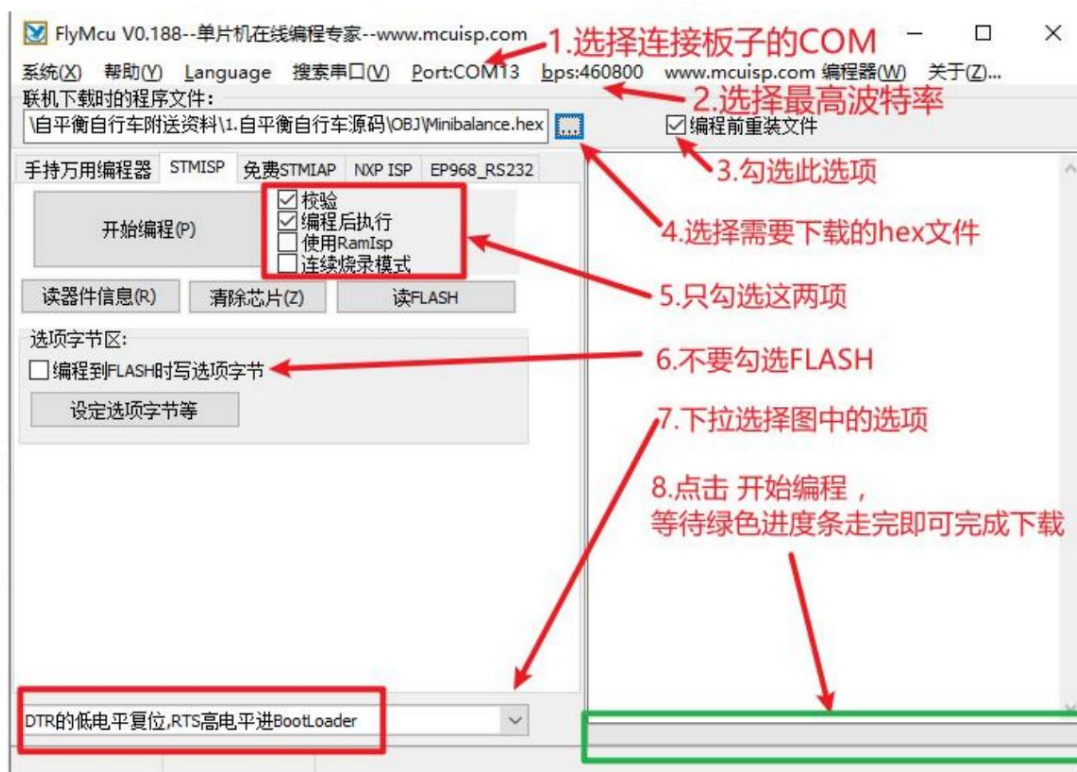


Figure 1-9 FlyMcu software setting details

OK, everything is ready, then click Start Programming, and the program will be downloaded!

Execute after programming, so after the program is downloaded, it will run automatically.